AI/BIgData 인텐시브 과정

강사 장철원

Section 0

코스소개

DAY1	DAY2	DAY3	DAY4	DAY5
데이터 분석을 위한 기초 수학			데이터 집계 및 시각화	
DAY6	**DAY7**	**DAY8**	**DAY9**	**DAY10**
	미니 프로젝트	확률분포, AB테스트		
DAY11	**DAY12**	**DAY13**	**DAY14**	**DAY15**
머신러닝				
DAY16	**DAY17**	**DAY18**	**DAY19**	**DAY20**
딥러닝 맛보기		파이널 프로젝트		

❑ 크로스밸리데이션

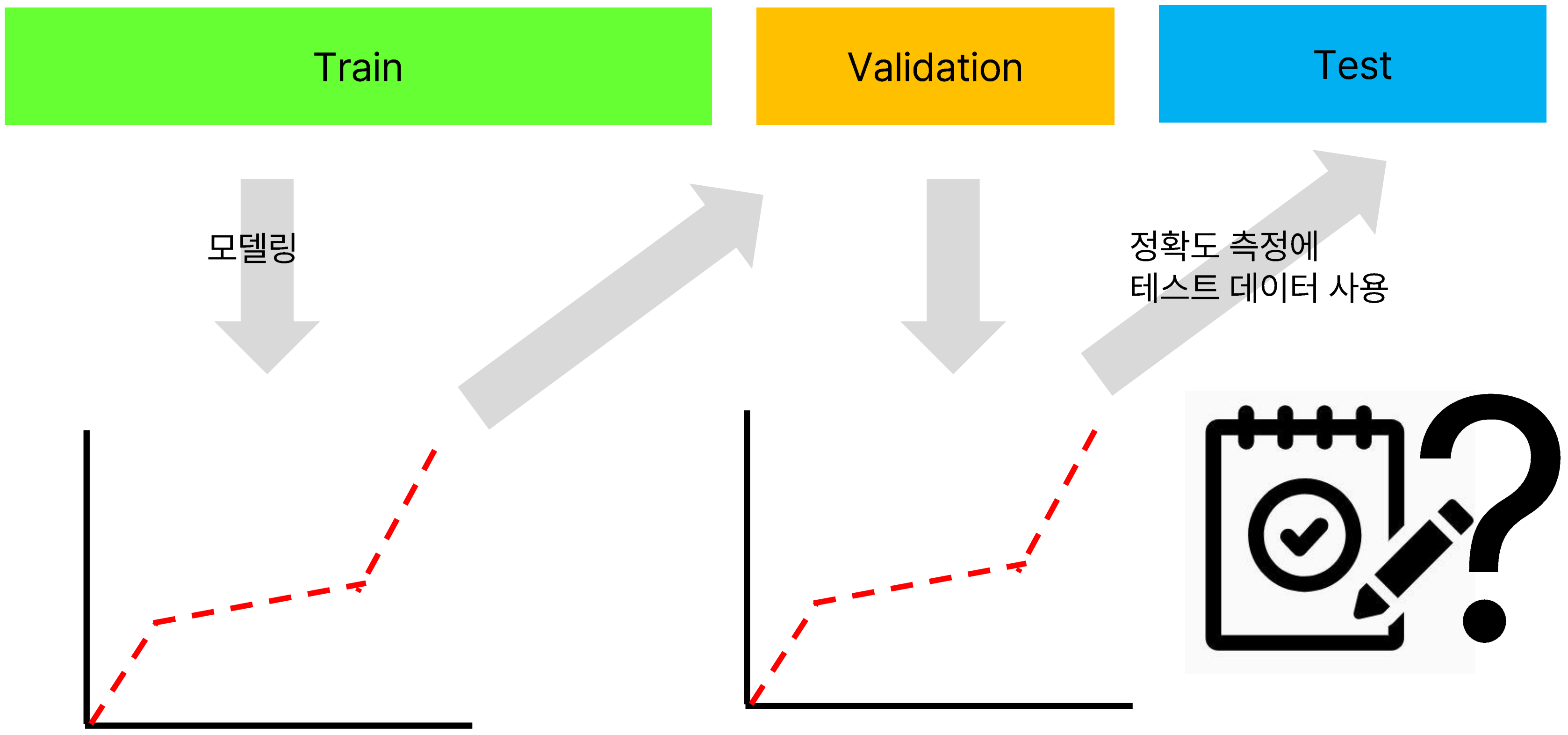
AI/BIgData 인텐시브 과정

Section 1. 크로스 밸리데이션

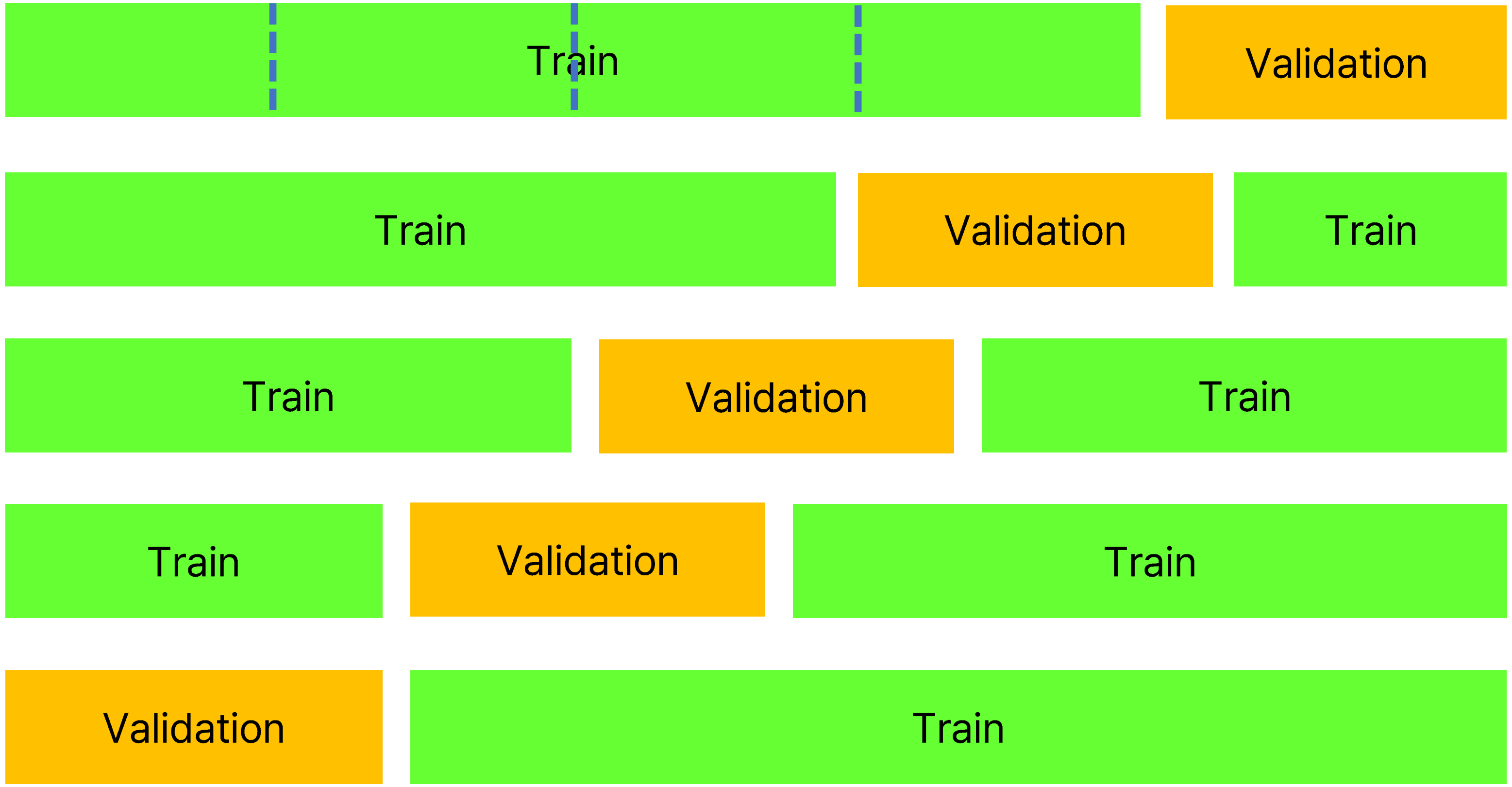
Section 1-1. 크로스 밸리데이션 분류

교차 검증(cross validation)

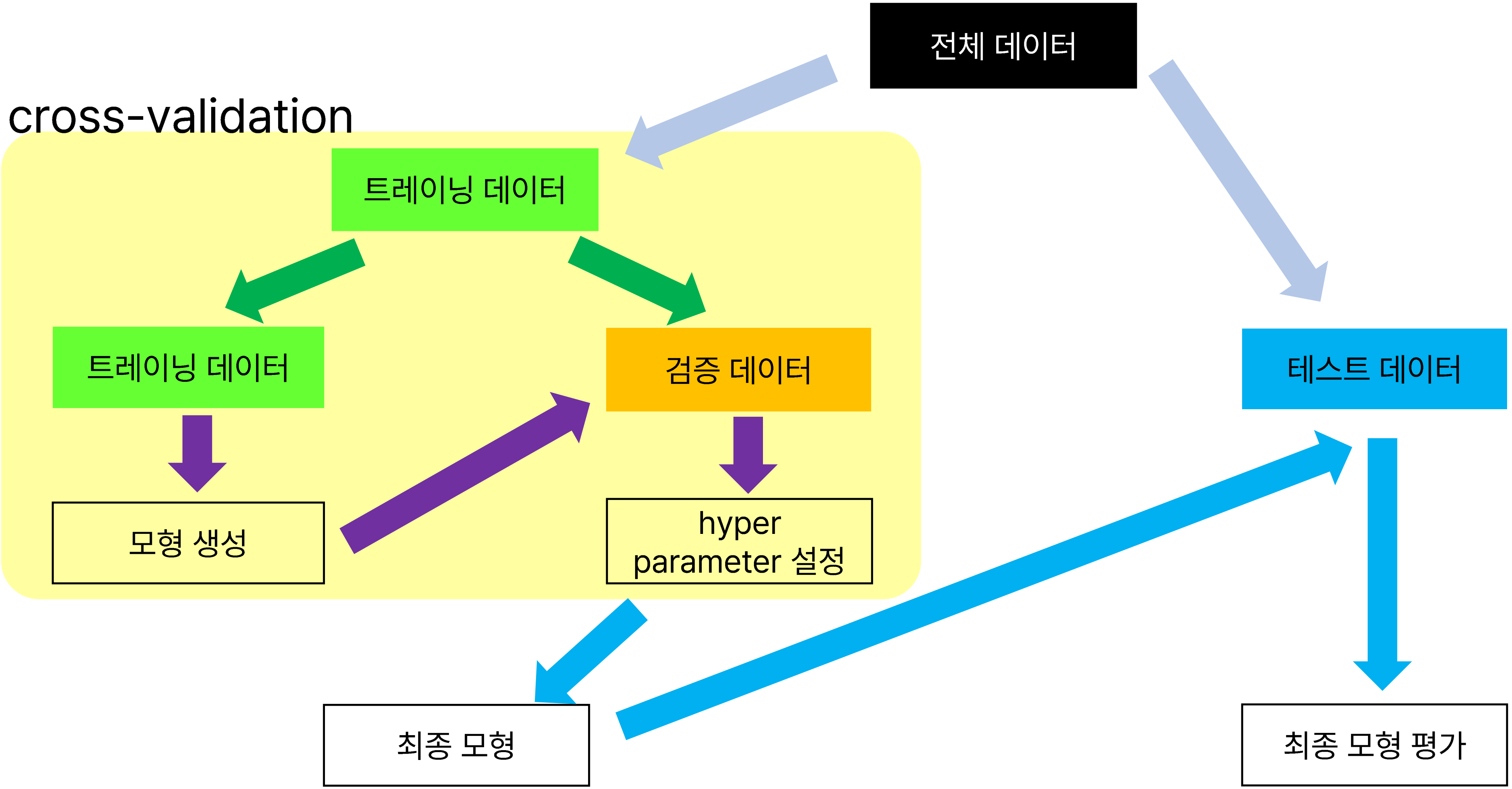
트레이닝 데이터 / 밸리데이션 데이터 / 테스트 데이터 분리



k fold cross validation



교차 검증(cross validation)



크로스 밸리데이션

In [1]:

```
import pandas as pd
df = pd.read_csv("./data/wine_data.csv")
```

In [2]:

```
features = ['Alcohol', 'Malic', 'Ash', 'Alcalinity', 'Magesium', 'Phenols', 'Flavanoids',
            'Nonflavanoids', 'Proanthocyanins', 'Color','Hue', 'Dilution', 'Proline']

X = df[features]
y = df['class']
```

크로스 밸리데이션

In [3]:
```
# 트레이닝/테스트 데이터 분할
from sklearn.model_selection import train_test_split
X_tn, X_te, y_tn, y_te=train_test_split(X,y,random_state=0)
```

In [4]:
```
# 데이터 표준화
from sklearn.preprocessing import StandardScaler
std_scale = StandardScaler()
std_scale.fit(X_tn)
X_tn_std = std_scale.transform(X_tn)
X_te_std  = std_scale.transform(X_te)
```

크로스 밸리데이션

In [5]:

```python
# 그리드 서치 학습
from sklearn import svm
from sklearn.model_selection import StratifiedKFold
from sklearn.model_selection import GridSearchCV

param_grid= {'kernel': ('linear', 'rbf'),
             'C': [0.5, 1, 10, 100]}
kfold = StratifiedKFold(n_splits=5, shuffle=True, random_state=0)
svc = svm.SVC(random_state=0)
grid_cv = GridSearchCV(svc, param_grid, cv=kfold, scoring='accuracy')
grid_cv.fit(X_tn_std, y_tn)
```

Out[5]:

```
▸ GridSearchCV
▸ estimator: SVC
    ▸ SVC
```

In [6]:
```python
# 그리드 서치 결과 확인
grid_cv.cv_results_
```

Out[6]:
```
{'mean_fit_time': array([0.00073509, 0.00103574, 0.00263205, 0.0008553 , 0.00104203,
        0.00121851, 0.00037541, 0.00083108]),
 'std_fit_time': array([0.00093357, 0.00087752, 0.00366261, 0.00104792, 0.00090734,
        0.00243702, 0.00046287, 0.00122318]),
 'mean_score_time': array([0.00141363, 0.00070448, 0.00013232, 0.00017943, 0.00028906,
        0.00015235, 0.00131979, 0.00119324]),
 'std_score_time': array([0.00237844, 0.00098571, 0.00026464, 0.00035887, 0.00043975,
        0.0003047 , 0.00136025, 0.00068336]),
 'param_C': masked_array(data=[0.5, 0.5, 1, 1, 10, 10, 100, 100],
              mask=[False, False, False, False, False, False, False, False],
        fill_value='?',
             dtype=object),
 'param_kernel': masked_array(data=['linear', 'rbf', 'linear', 'rbf', 'linear', 'rbf',
                    'linear', 'rbf'],
              mask=[False, False, False, False, False, False, False, False],
        fill_value='?',
             dtype=object),
 'params': [{'C': 0.5, 'kernel': 'linear'},
  {'C': 0.5, 'kernel': 'rbf'},
  {'C': 1, 'kernel': 'linear'},
  {'C': 1, 'kernel': 'rbf'},
  {'C': 10, 'kernel': 'linear'},
  {'C': 10, 'kernel': 'rbf'},
  {'C': 100, 'kernel': 'linear'},
  {'C': 100, 'kernel': 'rbf'}],
 'split0_test_score': array([0.88888889, 0.96296296, 0.88888889, 0.92592593, 0.88888889,
        0.92592593, 0.88888889, 0.92592593]),
 'split1_test_score': array([0.96296296, 1.        , 0.96296296, 0.96296296, 0.96296296,
        0.96296296, 0.96296296, 0.96296296]),
```

In [7]:

```python
# 그리드 서치 결과 확인(데이터프레임)

import numpy as np
import pandas as pd

np.transpose(pd.DataFrame(grid_cv.cv_results_))
```

	0	1	2	3	4	5	6	7
mean_fit_time	0.000735	0.001036	0.002632	0.000855	0.001042	0.001219	0.000375	0.000831
std_fit_time	0.000934	0.000878	0.003663	0.001048	0.000907	0.002437	0.000463	0.001223
mean_score_time	0.001414	0.000704	0.000132	0.000179	0.000289	0.000152	0.00132	0.001193
std_score_time	0.002378	0.000986	0.000265	0.000359	0.00044	0.000305	0.00136	0.000683
param_C	0.5	0.5	1	1	10	10	100	100
param_kernel	linear	rbf	linear	rbf	linear	rbf	linear	rbf
params	{'C': 0.5, 'kernel': 'linear'}	{'C': 0.5, 'kernel': 'rbf'}	{'C': 1, 'kernel': 'linear'}	{'C': 1, 'kernel': 'rbf'}	{'C': 10, 'kernel': 'linear'}	{'C': 10, 'kernel': 'rbf'}	{'C': 100, 'kernel': 'linear'}	{'C': 100, 'kernel': 'rbf'}
split0_test_score	0.888889	0.962963	0.888889	0.925926	0.888889	0.925926	0.888889	0.925926
split1_test_score	0.962963	1.0	0.962963	0.962963	0.962963	0.962963	0.962963	0.962963
split2_test_score	0.925926	0.962963	0.925926	0.962963	0.925926	0.962963	0.925926	0.962963
split3_test_score	1.0	0.961538	1.0	0.961538	1.0	0.961538	1.0	0.961538
split4_test_score	0.846154	1.0	0.846154	1.0	0.846154	1.0	0.846154	1.0
mean_test_score	0.924786	0.977493	0.924786	0.962678	0.924786	0.962678	0.924786	0.962678
std_test_score	0.054014	0.018384	0.054014	0.023431	0.054014	0.023431	0.054014	0.023431
rank_test_score	5	1	5	2	5	2	5	2

크로스 밸리데이션

In [8]:
```python
# 베스트 스코어
grid_cv.best_score_
```
Out[8]:
```
0.9774928774928775
```

In [9]:
```python
# 베스트 하이퍼파라미터
grid_cv.best_params_
```
Out[9]:
```
{'C': 0.5, 'kernel': 'rbf'}
```

In [10]:
```python
# 최종 모형
clf = grid_cv.best_estimator_
print(clf)
```
```
SVC(C=0.5, random_state=0)
```

크로스 밸리데이션

In [11]:

```python
# 크로스 밸리데이션 스코어 확인(1)
from sklearn.model_selection import cross_validate
metrics = ['accuracy', 'precision_macro', 'recall_macro', 'f1_macro']
cv_scores = cross_validate(clf, X_tn_std, y_tn,
                           cv=kfold, scoring=metrics)
cv_scores
```

Out[11]:

```
{'fit_time': array([0.        , 0.00316906, 0.        , 0.00152636, 0.00160766]),
 'score_time': array([0.00351453, 0.00155187, 0.00566816, 0.00306392, 0.0028975 ]),
 'test_accuracy': array([0.96296296, 1.        , 0.96296296, 0.96153846, 1.        ]),
 'test_precision_macro': array([0.96296296, 1.        , 0.96969697, 0.96969697, 1.        ]),
 'test_recall_macro': array([0.96666667, 1.        , 0.96296296, 0.95833333, 1.        ]),
 'test_f1_macro': array([0.9628483 , 1.        , 0.96451914, 0.96190476, 1.        ])}
```

In [12]:

```python
# 크로스 밸리데이션 스코어 확인(2)
from sklearn.model_selection import cross_val_score
cv_score = cross_val_score(clf, X_tn_std, y_tn,
                           cv=kfold, scoring='accuracy')
print(cv_score)
print(cv_score.mean())
print(cv_score.std())
```

```
[0.96296296 1.         0.96296296 0.96153846 1.        ]
0.9774928774928775
0.01838434849561446
```

크로스 밸리데이션

In [13]:

```
# 예측
pred_svm = clf.predict(X_te_std)
print(pred_svm)
```

```
[0 2 1 0 1 1 0 2 1 1 2 2 0 1 2 1 0 0 1 0 1 0 0 1 1 1 1 1 1 2 0 0 1 0 0 0 2
 1 1 2 0 0 1 1 1]
```

In [14]:

```
# 정확도
from sklearn.metrics import accuracy_score
accuracy = accuracy_score(y_te, pred_svm)
print(accuracy)
```

```
1.0
```

크로스 밸리데이션

In [15]:

```
# confusion matrix 확인
from sklearn.metrics import confusion_matrix
conf_matrix = confusion_matrix(y_te, pred_svm)
print(conf_matrix)
```

```
[[16  0  0]
 [ 0 21  0]
 [ 0  0  8]]
```

In [16]:

```
# 분류 레포트 확인
from sklearn.metrics import classification_report
class_report = classification_report(y_te, pred_svm)
print(class_report)
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        16
           1       1.00      1.00      1.00        21
           2       1.00      1.00      1.00         8

    accuracy                           1.00        45
   macro avg       1.00      1.00      1.00        45
weighted avg       1.00      1.00      1.00        45
```

AI/BIgData 인텐시브 과정

Section 1. 크로스 밸리데이션

Section 1-1. 크로스 밸리데이션 예측

크로스 밸리데이션

In [2]:
```
import pandas as pd
df = pd.read_csv("./data/house_prices.csv")
```

In [3]:
```
features = ['CRIM', 'ZN', 'INDUS', 'CHAS', 'NOX', 'RM', 'AGE', 'DIS', 'RAD', 'TAX', 'PTRATIO', 'B', 'LSTAT']

X = df[features]
y = df['MEDV']
```

In [4]:
```
# 트레이닝/테스트 데이터 분할
from sklearn.model_selection import train_test_split
X_tn, X_te, y_tn, y_te=train_test_split(X,y,random_state=0)
```

In [5]:
```
# 데이터 표준화
from sklearn.preprocessing import StandardScaler
std_scale = StandardScaler()
std_scale.fit(X_tn)
X_tn_std = std_scale.transform(X_tn)
X_te_std  = std_scale.transform(X_te)
```

크로스 밸리데이션

In [10]:

```python
# 그리드 서치 학습
from sklearn import svm
from sklearn.model_selection import KFold
from sklearn.model_selection import GridSearchCV

param_grid= {'kernel': ('linear', 'rbf'),
             'C': [0.5, 1, 10, 100]}
kfold = KFold(n_splits=5, shuffle=True, random_state=0)
svr = svm.SVR()
grid_cv = GridSearchCV(svr, param_grid, cv=kfold, scoring='neg_mean_squared_error')
grid_cv.fit(X_tn_std, y_tn)
```

Out[10]:

```
▸ GridSearchCV
▸ estimator: SVR
    ▸ SVR
```

크로스 밸리데이션 실습

In [11]:

```python
# 그리드 서치 결과 확인(데이터프레임)

import numpy as np
import pandas as pd

np.transpose(pd.DataFrame(grid_cv.cv_results_))
```

Out[11]:

	0	1	2	3	4	5	6	7
mean_fit_time	0.004308	0.003131	0.004675	0.00253	0.015536	0.005196	0.096247	0.013163
std_fit_time	0.000869	0.001588	0.001412	0.001406	0.003971	0.004069	0.038219	0.000805
mean_score_time	0.000587	0.001271	0.000398	0.001447	0.000823	0.000783	0.00058	0.001466
std_score_time	0.000484	0.000721	0.000488	0.001005	0.000702	0.000744	0.000539	0.000526
param_C	0.5	0.5	1	1	10	10	100	100
param_kernel	linear	rbf	linear	rbf	linear	rbf	linear	rbf
params	{'C': 0.5, 'kernel': 'linear'}	{'C': 0.5, 'kernel': 'rbf'}	{'C': 1, 'kernel': 'linear'}	{'C': 1, 'kernel': 'rbf'}	{'C': 10, 'kernel': 'linear'}	{'C': 10, 'kernel': 'rbf'}	{'C': 100, 'kernel': 'linear'}	{'C': 100, 'kernel': 'rbf'}
split0_test_score	-17.445032	-36.17043	-17.386875	-27.228367	-17.458346	-17.409092	-17.446742	-12.562522
split1_test_score	-18.593747	-16.916761	-18.521081	-11.816775	-18.665759	-5.996846	-18.667153	-9.235812
split2_test_score	-40.571874	-76.712963	-38.982199	-57.052392	-37.459318	-26.609204	-37.063074	-15.662599
split3_test_score	-23.262262	-34.792208	-23.221267	-24.185331	-23.007632	-9.182264	-22.978231	-6.061032
split4_test_score	-17.088043	-33.352135	-17.18929	-23.301751	-17.192183	-9.34963	-17.19154	-7.405381
mean_test_score	-23.392191	-39.588899	-23.060142	-28.716923	-22.756648	-13.709407	-22.669348	-10.185469
std_test_score	8.868985	19.82833	8.255832	15.103569	7.642017	7.470993	7.490986	3.501173
rank_test_score	6	8	5	7	4	2	3	1

크로스 밸리데이션

In [12]:
```
# 베스트 스코어
grid_cv.best_score_
```

Out[12]:
```
-10.185469344281149
```

In [13]:
```
# 베스트 하이퍼파라미터
grid_cv.best_params_
```

Out[13]:
```
{'C': 100, 'kernel': 'rbf'}
```

In [14]:
```
# 최종 모형
model = grid_cv.best_estimator_
print(model)
```

```
SVR(C=100)
```

In [15]:
```
# 예측
pred_svm = model.predict(X_te_std)
print(pred_svm)
```

```
[22.36813301 22.68383242 24.87514322 10.71088583 20.84027476 19.73537584
 23.616982   19.95204534 20.95058207 22.88723384 18.0129821  10.34899842
 14.71816571  5.72642875 47.13543116 31.19225615 23.92516254 36.38402412
 29.8542525  21.3802061  23.4583459  19.83747957 20.28994301 27.14446529
 20.76220706 21.44860078 16.90027694 17.73675701 37.75599541 18.56694782
```

크로스 밸리데이션

In [16]:

```python
# 모형 평가-MSE
from sklearn.metrics import mean_squared_error
mse = mean_squared_error(y_te, pred_svm)
print(mse)
```

```
17.51711113685591
```

감사합니다.

Q & A